ARJUNA JEE 2027

Lecture -10 | Mathematics

SEQUENCE & SERIES

By- Sachin Jakhar Sir



Topics *to be covered*

A **Method of Subtraction**

B **Telescopic Method of Difference** → TMOD.

C **Question Practice**

D



DIBY SOLUTIONS

Chapters | Study Material | Digital Books

SM - 01
PYQ Practice Sheet || Only PDF
34 Notes

SM - 02
Short Notes || Only PDF
25 Notes

SM - 03
Mind Maps || Only PDF
16 Notes

SM - 04
PYQ's Blueprint || Only PDF
17 Notes

SM - 05
Bridge Course Lectures
1 Lectures • 1 Notes

SM - 06
Concise Summary Notes || Onl...
51 Notes

SM - 07
DIBY By Sachin Sir || Only PDF
5 Notes

SM - 08
PYQ Practice Sheet
4 DPP • 4 Notes

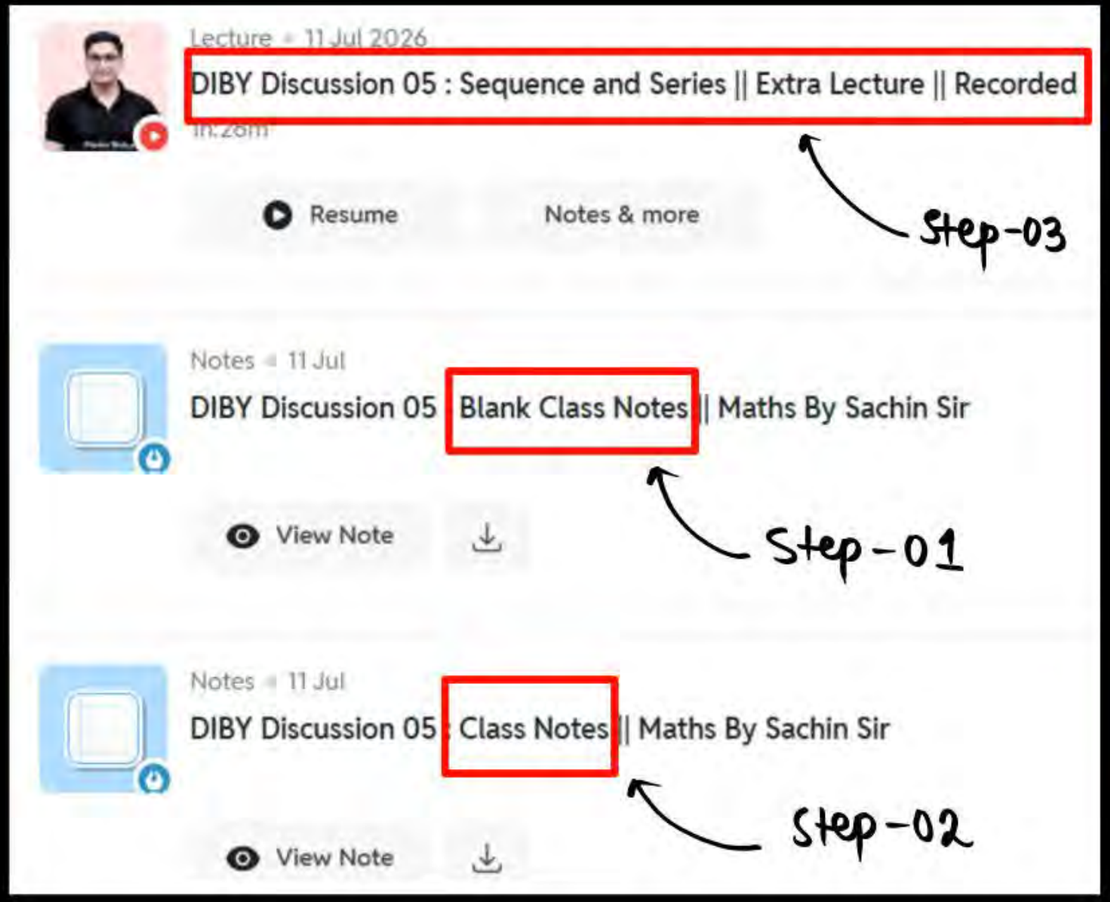
SM - 09
Mission 11th Comeback
1 Lectures • 1 Notes

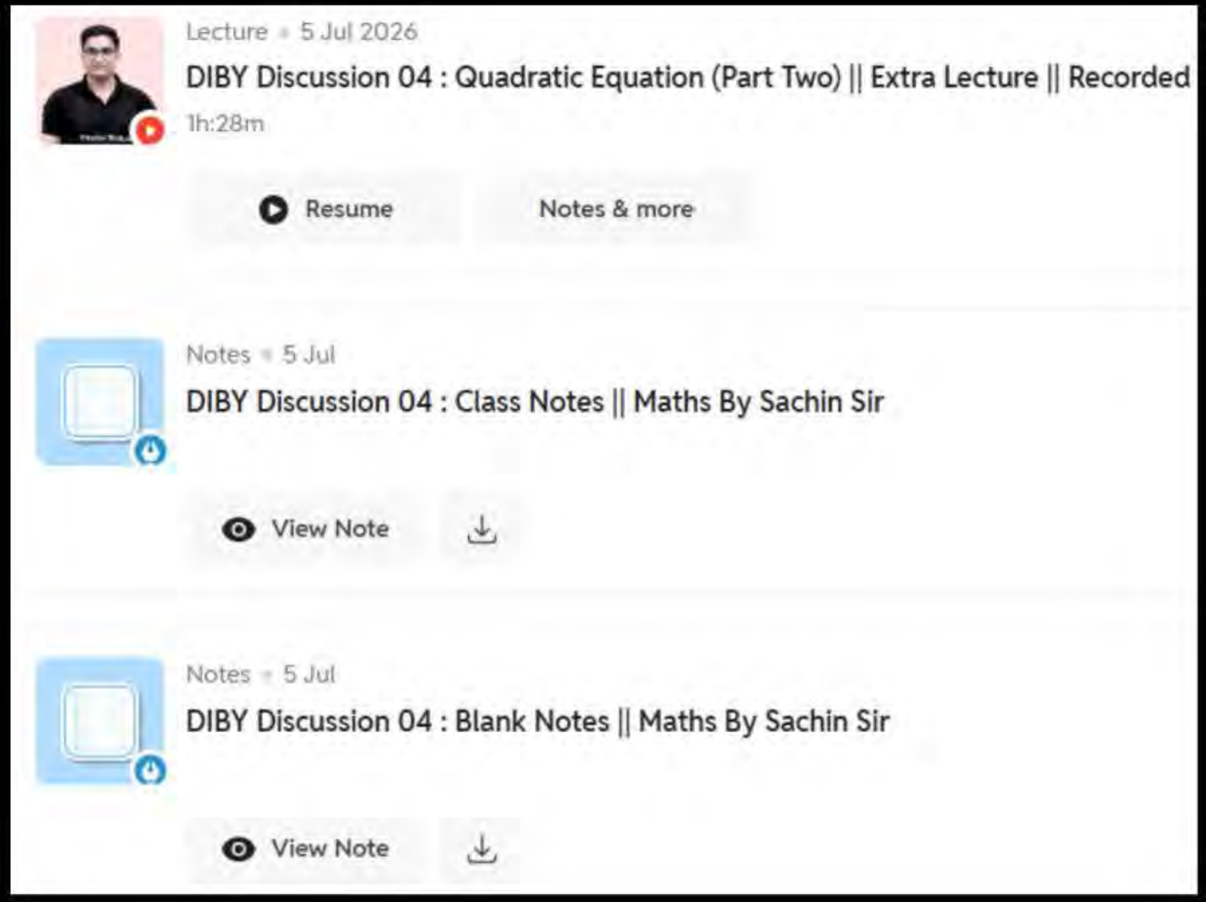
SM - 10
DIBY Discussion By Sachin Sir
5 Lectures • 10 Notes

SM - 11
Handwritten Notes || Only PDF
4 Notes

SM - 12
JEE Replica Sheet
4 DPP • 4 Notes

DIBY SOLUTIONS





Question Practice

Acidic Question on Multiple Sigma

Compute the value of : $S = \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \sum_{i=0}^{\infty} \frac{1}{2^i \cdot 3^j 5^k}$

$$S = \sum_{K=0}^{\infty} \sum_{j=0}^{\infty} \left(\sum_{i=0}^{\infty} \frac{1}{2^i 3^j 5^k} \right)$$

$$\left(\frac{1}{2^0 3^j 5^k} + \frac{1}{2^1 3^j 5^k} + \frac{1}{2^2 3^j 5^k} + \dots \infty \right) = \frac{1}{3^j 5^k} \left(1 + \frac{1}{2^1} + \frac{1}{2^2} + \dots \infty \right)$$

$$\frac{1}{1 - \frac{1}{2}} = 2$$

$$= 2 \cdot \frac{1}{3^j} \frac{1}{5^k}$$

Acidic Question on Multiple Sigma

Compute the value of : $S = \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \sum_{i=0}^{\infty} \frac{1}{2^i \cdot 3^j 5^k}$

$$S = \sum_{k=0}^{\infty} \left(\sum_{j=0}^{\infty} \left(2 \cdot \frac{1}{3^j} \cdot \frac{1}{5^k} \right) \right) = 2 \times \frac{1}{5^k} \left(\frac{1}{3^0} + \frac{1}{3^1} + \frac{1}{3^2} + \dots \infty \right)$$

$a = 1, \; r = \frac{1}{3}$

$$\frac{1}{1 - \frac{1}{3}} = \frac{1}{\left(\frac{2}{3}\right)} = \frac{3}{2}$$

$$= 2 \times \frac{1}{5^k} \times \frac{3}{2} = \frac{3}{5^k}$$

Acidic Question on Multiple Sigma

Compute the value of : $S = \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \sum_{i=0}^{\infty} \frac{1}{2^i . 3^j 5^k}$

$$S = \sum_{K=0}^{\infty} \left(\frac{3}{5^K}\right) = \frac{3}{5^0} + \frac{3}{5^1} + \frac{3}{5^2} + \cdots \infty$$

$$= 3\left(1 + \frac{1}{5} + \frac{1}{5^2} + \cdots \infty\right)$$

$$= 3 \times \left(\frac{1}{1-\frac{1}{5}}\right) = 3 \times \frac{1}{\left(\frac{4}{5}\right)} = \frac{3 \times 5}{4}$$ fot.

Challenger Question [JEE Mains-2022]

82%

If $A = \sum_{n=1}^{\infty} \frac{1}{(3+(-1)^n)^n}$ and $B = \sum_{n=1}^{\infty} \frac{(-1)^n}{(3+(-1)^n)^n}$, then $\frac{A}{B}$ is equal to :

A $\frac{11}{9}$

B 1

C $-\frac{11}{9}$

D $-\frac{11}{3}$

$$B = \frac{-1}{2^1} + \frac{1}{4^2} - \frac{1}{2^3} + \frac{1}{4^4} - \frac{1}{2^5} + \frac{1}{4^6} - \cdots \infty$$

$$A = \frac{1}{(3+(-1)^1)^1} + \frac{1}{(3+(-1)^2)^2} + \frac{1}{(3+(-1)^3)^3} + \frac{1}{(3+(-1)^4)^4} + \cdots \infty$$

$$A = \frac{1}{2^1} + \frac{1}{4^2} + \frac{1}{2^3} + \frac{1}{4^4} + \cdots \infty = \left(\frac{1}{2^1} + \frac{1}{2^3} + \frac{1}{2^5} + \cdots \infty\right) + \left(\frac{1}{4^2} + \frac{1}{4^4} + \frac{1}{4^6} + \cdots \infty\right)$$

$a = \frac{1}{2}, r = \frac{1}{2^2}$ $\qquad$ $a = \frac{1}{4^2}, r = \frac{1}{4^2}$

Question [JEE Mains-2019]

The sum $1+\frac{1^3+2^3}{1+2}+\frac{1^3+2^3+3^3}{1+2+3}+\ldots+\frac{1^3+2^3+3^3+\ldots+15^3}{1+2+3+\ldots+15}-\frac{1}{2}(1+2+3+\ldots+15)$ is equal to

Ans: B

A 1860

B 620

C 660

D 1240

Question [JEE Mains-2025]

[Ans. : (C)]

If $\frac{1}{1^4}+\frac{1}{2^4}+\frac{1}{3^4}+\cdots\infty=\frac{\pi^4}{90}$, (use)

$\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots\infty=\alpha$,

$\frac{1}{2^4}+\frac{1}{4^4}+\frac{1}{6^4}+\cdots\infty=\beta$, then $\frac{\alpha}{\beta}$ is equal to

$\frac{\alpha}{\beta}=\frac{\frac{15}{16}\times\frac{\pi^4}{90}}{\frac{1}{16}\times\frac{\pi^4}{90}}=15$

$2^4=(2\times1)^4=2^4\cdot1^4$

$4^4=(2\times2)^4=2^4\cdot2^4$

$6^4=(2\times3)^4=2^4\cdot3^4$

$\beta=\frac{1}{2^4}+\frac{1}{4^4}+\frac{1}{6^4}+\frac{1}{8^4}+\ldots\infty$

$=\frac{1}{2^4}\left(\frac{1}{1^4}+\frac{1}{2^4}+\frac{1}{3^4}+\frac{1}{4^4}+\ldots\infty\right)$

$\beta=\frac{1}{16}\times\frac{\pi^4}{90}$

$\alpha=\left(\frac{1}{1^4}+\frac{1}{2^4}+\frac{1}{3^4}+\frac{1}{4^4}+\frac{1}{5^4}+\frac{1}{6^4}+\frac{1}{7^4}+\frac{1}{8^4}\ldots\right)-\left(\frac{1}{2^4}+\frac{1}{4^4}+\frac{1}{6^4}+\frac{1}{8^4}\ldots\right)$

$\alpha=\frac{\pi^4}{90}-\beta$ (β)

$\alpha=\frac{\pi^4}{90}-\frac{1}{16}\times\frac{\pi^4}{90}=\frac{\pi^4}{90}\left(\frac{15}{16}\right)$

A	23
B	18
C	15
D	14

Challenger Question

If $\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\cdots\infty=\frac{\pi^2}{6}$, then choose correct option -

"Multiple Correct"
new
(80% DIBY)

A $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\cdots\infty=\frac{\pi^2}{8}$

B $\frac{1}{1^2}-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+\cdots\infty=\frac{\pi^2}{12}$

C $\frac{1}{4^2}+\frac{1}{8^2}+\frac{1}{12^2}+\cdots\infty=\frac{\pi^2}{96}$

D $\frac{1}{2^2}+\frac{1}{4^2}+\frac{1}{6^2}+\cdots\infty=\frac{\pi^2}{24}$

METHOD OF SUBTRACTION TO WRITE T_n

$T_n = 4n^2 - 1$

$n=1 \Rightarrow T_1 = 3$

$n=2 \Rightarrow T_2 = 15$

$n=3 \Rightarrow T_3 = 35$ ...

Question

Find the sum to n terms of the series $3 + 15 + 35 + 63 + ...$

$S = 3 + 15 + 35 + 63 + ... + T_{n-1} + T_n$

$S = 0 + 3 + 15 + 35 + ... + T_{n-1} + T_n$

$\ominus$

$0 = 3 + 12 + 20 + 28 + ... (\ ) + (\ ) - T_n$

$+8 \quad +8 \quad +8$

A.P. $(a = 12, d = 8, \text{n.o.t.} = n-1)$

$$T_n = 3 + \frac{n-1}{2}\left(2 \times 12 + (n-1-1)8\right)$$

$$T_n = 3 + \frac{n-1}{2}\left(2\times 12 + \overbrace{(n-1-1)}^{n-2}8\right)$$

$$= 3 + \frac{(n-1)}{2}(24 + 8n - 16)$$

$$= 3 + \frac{(n-1)}{2}(8n+8)$$

$$= 3 + \frac{(n-1)}{\cancel{2}}\overset{4}{\cancel{8}}(n+1)$$

$$= 3 + 4(n^2-1) = 3 + 4n^2 - 4$$

$$= 4n^2 - 1$$

$$\boxed{T_n = 4n^2 - 1}$$

$$S = \sum T_n$$

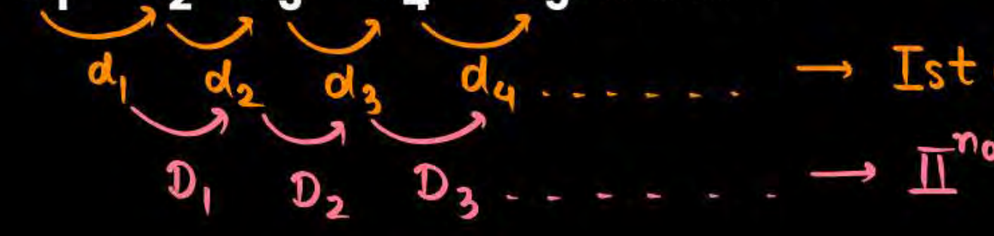
$$S = \sum_{n=1}^{n}(4n^2-1)$$

$$= 4\sum_{n=1}^{n} n^2 - \sum_{n=1}^{n} 1$$

$$S = \left(4\times\frac{n(n+1)(2n+1)}{6} - n\times 1\right)$$ fnf.

SHORTCUT METHOD TO FIND T_n

Sequence : $a_1 \quad a_2 \quad a_3 \quad a_4 \quad a_5 \ldots \ldots \ldots$

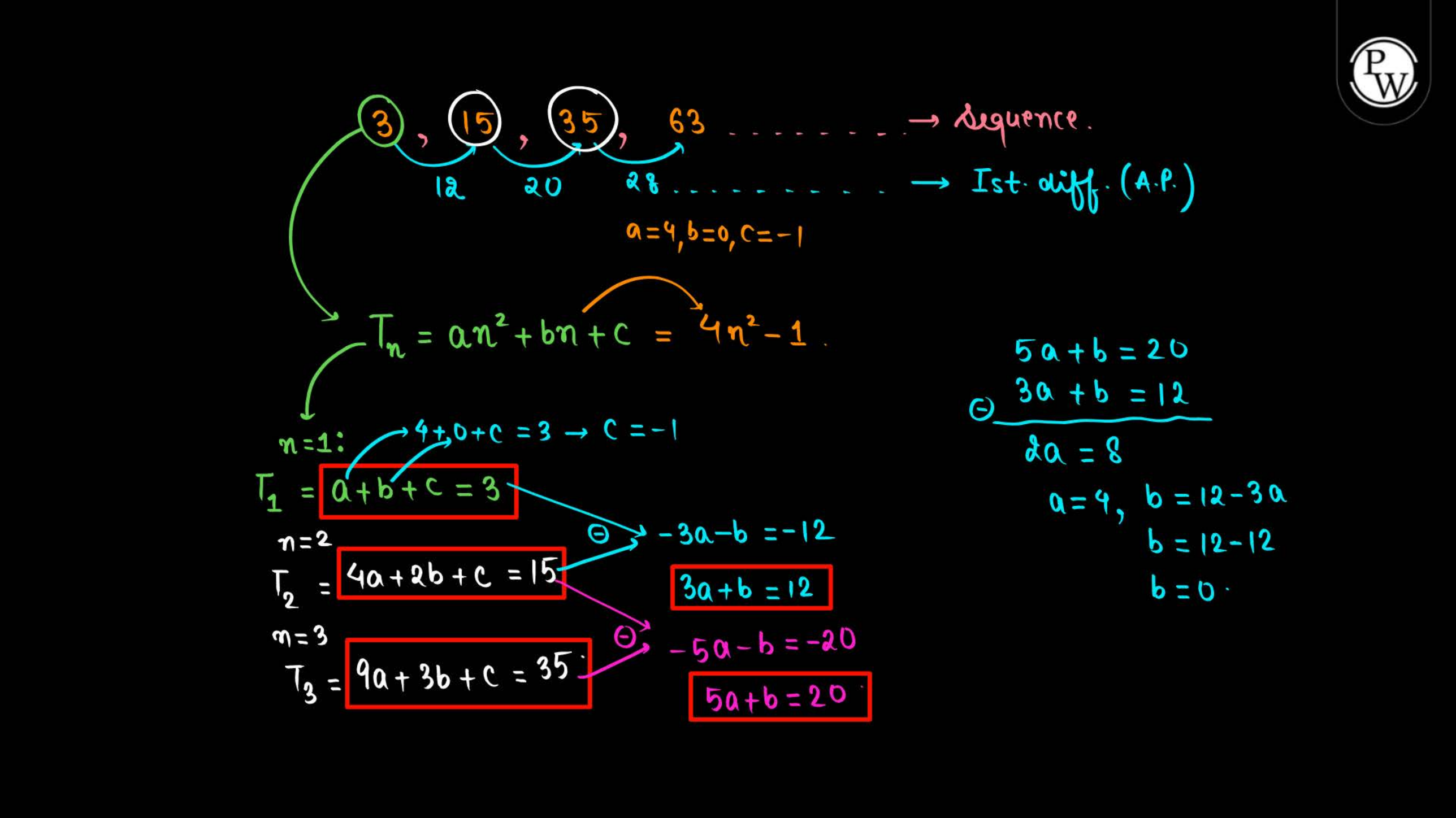


If Ist – difference is in A.P., then $T_n = an^2 + bn + c$ ★★★★★

If IInd – difference is in A.P., then $T_n = an^3 + bn^2 + cn + d$

If Ist – difference is in G.P., then $T_n = a(C.R.)^n + bn + c$ ★★★★★

If IInd – difference is in G.P., then $T_n = a(C.R.)^n + bn^2 + cn + d$


3, 15, 35, 63 ..... → Sequence.
12 20 28 ..... → Ist diff. (A.P.)
a=4, b=0, c=-1
$T_n = an^2 + bn + c = 4n^2 - 1$.
n=1: 4+0+c=3 → c=-1
$T_1 = a+b+c = 3$
n=2
$T_2 = 4a+2b+c = 15$
n=3
$T_3 = 9a+3b+c = 35$
⊖ $-3a-b = -12$
$3a+b = 12$
⊖ $-5a-b = -20$
$5a+b = 20$
$5a+b = 20$
⊖ $3a+b = 12$
$2a = 8$
$a = 4$, $b = 12-3a$
$b = 12-12$
$b = 0$.

Question

$2 + 5 + 14 + 41 + 122 + \ldots\ldots$ upto n terms.

3 9 27 81 . . . . . → Ist diff G.P. with CR = 3.

$T_n = a(3)^n + bn + c$ → $a = \frac{1}{2}, b = 0, c = \frac{1}{2}$

$n = 1$:

$T_1 = 3a + b + c = 2$ → $3\left(\frac{1}{2}\right) + 0 + c = 2$

$c = 2 - \frac{3}{2} = \frac{1}{2}$

$n = 2$:

$T_2 = 9a + 2b + c = 5$

$n = 3$

$T_3 = 27a + 3b + c = 14$

$\ominus$ $-6a - b = -3$ → $6a + b = 3$ ✓

$\ominus$ $-18a - b = -9$ → $18a + b = 9$

$-12a = -6$

$a = \frac{1}{2}$, $b = 3 - 6a$

$= 3 - 6 \times \frac{1}{2} = 0$.

2, 5, 14, 41, 122 .....

$T_n = a(3)^n + bn + c$ $\rightarrow a = \frac{1}{2}, b = 0, c = \frac{1}{2}$

$T_n = \frac{1}{2}(3)^n + 0 + \frac{1}{2} = \boxed{\frac{3^n + 1}{2}}$

$n = 1$ $\rightarrow T_1 = \frac{3^1 + 1}{2} = 2$

$n = 2$ $\rightarrow T_2 = \frac{3^2 + 1}{2} = \frac{10}{2} = 5$

$n = 3$ $\rightarrow T_3 = \frac{3^3 + 1}{2} = \frac{28}{2} = 14$

2, 5, 14, 41, 122 .....

$T_n = \boxed{\frac{3^n+1}{2}}$

$S = 2 + 5 + 14 + 41 + 122 + \ldots$

$S = \sum_{n=1}^{n} \frac{3^n+1}{2} = \frac{1}{2}\left(\sum_{n=1}^{n} 3^n + \sum_{n=1}^{n} 1\right)$

$= \frac{1}{2}\left\{\left(3^1+3^2+3^3+\ldots+3^n\right) + n\times 1\right\}$

$S = \frac{1}{2}\left(3\left(\frac{3^n-1}{3-1}\right)+n\right)$ fnf.

Question

Find the sum of the following series to n terms :

$5 + 7 + 13 + 31 + 85 + \ldots\ldots\ldots$



Question [JEE Mains-2023]

Let a_n be the nth term of the series $5 + 8 + 14 + 23 + 35 + 50 + \ldots\ldots$ and $S_n = \sum_{k=1}^{n} a_k$. Then $S_{30} - a_{40}$ equal to



A 11310

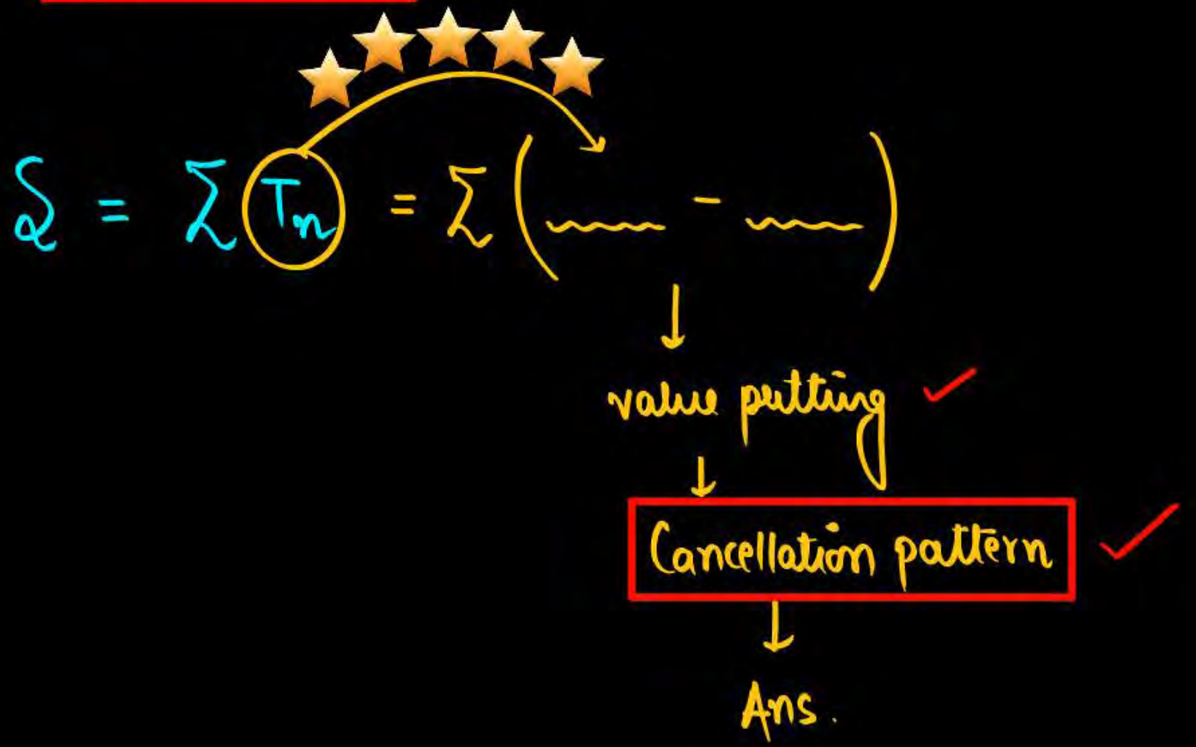
B 11280

C 11290

D 11260

TELESCOPIC METHOD OF DIFFERENCE (TMOD)

Also known as V_n Method



① $\sum_{r=1}^{5}\left((r+1)^2 - r^2\right)$ $= \cancel{2^2} - 1^2$

$+ \cancel{3^2} - \cancel{2^2}$

$+ \cancel{4^2} - \cancel{3^2}$

$+ \cancel{5^2} - \cancel{4^2}$

$+ \underline{6^2} - \cancel{5^2}$

$= 6^2 - 1^2$

$= 35.$

② $\sum_{r=1}^{7}\left((r+2)^2 - r^2\right)$ $= \cancel{3^2} - 1^2$

$+ \cancel{4^2} - 2^2$

$+ \cancel{5^2} - \cancel{3^2}$

$+ \cancel{6^2} - \cancel{4^2}$

$+ \cancel{7^2} - \cancel{5^2}$

$+ \cancel{8^2} - \cancel{6^2}$

$+ 9^2 - \cancel{7^2}$

Ans. $\Rightarrow 9^2 + 8^2 - 1^2 - 2^2$

TYPE-01: CONTINUOUS TERMS OF A.P. ARE IN MULTIPLICATION IN NUMERATOR OR DENOMINATOR

Question

Solve : $S = 1 \times 2 + 2 \times 3 + 3 \times 4 + \cdots$ **upto 15 Terms.**

M-I: $T_n = n(n+1)$

$$S = \sum_{n=1}^{15} (n^2+n) \checkmark$$

TYPE-01: CONTINUOUS TERMS OF A.P. ARE IN MULTIPLICATION IN NUMERATOR OR DENOMINATOR

Question

Solve : $S = 1 \times 2 + 2 \times 3 + 3 \times 4 + \cdots$ **upto 15 Terms.**

M-I ; T.H.O.D. $T_n = \frac{n(n+1)}{3}\left((n+2) - (n-1)\right)$

last Term ke agli term

first term ke pichli term

$$T_n = \frac{n(n+1)}{3}\left((n+2) - (n-1)\right)$$

$$T_n = \frac{n(n+1)}{3}\Big((n+2)-(n-1)\Big)$$

$$T_n = n(n+1) = \frac{n(n+1)(n+2)}{3} - \frac{(n-1)n(n+1)}{3}$$

$$T_1 = 1\times 2 = \frac{1\times 2\times 3}{3} - 0$$

$$T_2 = 2\times 3 = \frac{2\times 3\times 4}{3} - \frac{1\times 2\times 3}{3}$$

$$T_3 = 3\times 4 = \frac{3\times 4\times 5}{3} - \frac{2\times 3\times 4}{3}$$

$$\vdots$$

$$T_n = n(n+1) = \frac{n(n+1)(n+2)}{3} - \frac{(n-1)n(n+1)}{3}$$

add

$$S = \frac{n(n+1)(n+2)}{3}$$

$n=15$

$$S = \frac{15\times 16\times 17}{3}$$

Ques:-

$$S = \boxed{1\times2\times3\times4} + \boxed{2\times3\times4\times5} + \boxed{3\times4\times5\times6} + \dots\dots$$

$$T_n = \frac{n(n+1)(n+2)(n+3)}{5}\left(\boxed{(n+4)-(n-1)}\right)$$

$$= \frac{n(n+1)(n+2)(n+3)(n+4)}{5} - \frac{(n-1)n(n+1)(n+2)(n+3)}{5}$$

$$\boxed{S = \frac{n(n+1)(n+2)(n+3)(n+4)}{5}}$$

$$T_1 = \frac{1\times2\times3\times4\times5}{5} - 0$$

$$T_2 = \frac{2\times3\times4\times5\times6}{5} - \frac{1\times2\times3\times4\times5}{5}$$

$$\vdots$$

$$T_n = \frac{n(n+1)(n+2)(n+3)(n+4)}{5} - \frac{(n-1)n(n+1)(n+2)(n+3)}{5}$$

Ques:-

$S = 3\times5\times7 + 5\times7\times9 + 7\times9\times11 + \ldots$ upto n-Terms.

$$T_n = \frac{(2n+1)(2n+3)(2n+5)}{8}\left((2n+7)-(2n-1)\right)$$

$$T_n = \frac{(2n+1)(2n+3)(2n+5)(2n+7)}{8} - \frac{(2n-1)(2n+1)(2n+3)(2n+5)}{8}$$

add.

$T_1 =$

$T_2 =$

$T_3 =$

$\vdots$

$T_n =$

DO IT BY YOURSELF (DIBY)

PROPERTIES OF SIGMA

DIBY-74

Compute the sum of the series whose n^{th} term is given by $T_n = n(n+1)(3n-1)$.

(A) $\frac{n(n+1)(3n^2+3n-2)}{12}$

(B) $\frac{n(n+1)(9n^2+15n+2)}{12}$

(C) $\frac{n(n+1)(n+2)(9n-1)}{12}$

(D) $\frac{n(n+1)(3n^2+n+2)}{4}$

[Ans. : (C)]

DIBY-75

Compute the sum $(31)^2 + (32)^2 + (33)^2 + \cdots \ldots + (50)^2$.

(A) 32,925 (B) 33,470 (C) 34,185 (D) 35,310

[Ans. : (B)]

DIBY-76

The value(S) of the positive integer n for which the quadratic equation, $\sum_{k=1}^{n} (x+k-1)(x+k) = 10n$ has solutions α and $\alpha+1$ for some α is.

(A) 10 (B) 11 (C) 12 (D) 13

[Ans. : (B)]

PROPERTIES OF SIGMA

DIBY-77

The sum $\sum_{k=1}^{20}(1+2+3+\ldots..+k)$ is [JEE Mains-2020]

[Ans. : 1540]

DIBY-78

Let S_n be the sum to n-terms of an arithmetic progression 3, 7, 11, ... If $40 < \left(\frac{6}{n(n+1)}\sum_{k=1}^{n} S_k\right) < 42$, then n equals [JEE Mains-2024]

[Ans. : 9]

DIBY-79

The sum of the series $1+2\times3+3\times5+4\times7+\cdots$ upto 11th term is: [JEE Mins-2019]

(A) 915 (B) 946 (C) 945 (D) 916

[Ans. : (B)]

PROPERTIES OF SIGMA

DIBY-80

Let $S_K = \frac{1+2+\cdots+K}{K}$ and $\sum_{j=1}^{n} S_j^2 = \frac{n}{A}\left(Bn^2 + Cn + D\right)$ where $A, B, C, D \in \mathbb{N}$ and A has least value. Then [JEE Mains-2023]

(A) $A + C + D$ is not divisible by B

(B) $A + B = 5(D - C)$

(C) $A + B + D$ is divisible by 5

(D) $A + B$ is divisible by D

[Ans. : (D)]

DIBY-81

If $\frac{1^3+2^3+3^3+\cdots \text{ upto } n \text{ terms}}{1\cdot3+2\cdot5+3\cdot7+\cdots \text{ upto } n \text{ terms}} = \frac{9}{5}$, then the value of n is [JEE Mains-2023]

[Ans. : 5]

DIBY-82

If $\gcd(m, n) = 1$ and $1^2 - 2^2 + 3^2 - 4^2 + \cdots + (2021)^2 - (2022)^2 + (2023)^2 = 1012m^2n$, then $m^2 - n^2$ is equal to [JEE Mains-2023]

(A) 200 (B) 240 (C) 220 (D) 180

[Ans. : (B)]

PROPERTIES OF SIGMA

DIBY-83

If the sum of the first 15 terms of the series $\left(\frac{3}{4}\right)^3 + \left(1\frac{1}{2}\right)^3 + \left(2\frac{1}{4}\right)^3 + 3^3 + \left(3\frac{3}{4}\right)^3 + \cdots$ is equal to $225k$, then k is equal to: [JEE Mains-2019]

(A) 108 (B) 27 (C) 54 (D) 9

[Ans. : (C)]

WRITING T_n FOR SERIES WITH COMMON DIFFERENCE IN AP OR GP

DIBY-84

S_n = 2 + 5 + 14 + 41 + 122 + ... Upto n-terms. (done in class.)

[Ans. : $\frac{3^{n+1}+2n-3}{4}$]

DIBY-85

The sum $1 + 3 + 11 + 25 + 45 + 71 + \cdots$ upto 20 terms, is equal to

[JEE Mains-2025]

(A) 7240 (B) 7130 (C) 6982 (D) 8124

[Ans. : (A)]

DIBY-86

If $S_n = 4 + 11 + 21 + 34 + 50 + \cdots$ to n terms, then $\frac{1}{60}(S_{29} - S_9)$ [JEE Mains-2023]

(A) 226 (B) 220 (C) 223 (D) 227

[Ans. : (C)]

It is never about end result

It's always about Journey

Thank You

#futureIITians